Somjit Nath

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Research Interests

Reinforcement Learning (RL), Representation Learning, Model Based RL, Exploration in RL

Education

Mila & École de technologie supérieure (ÉTS) Montréal, Canada
Ph.D. in Engineering; *Jan 2022 – Dec 2025 (Expected)*
Advisor: Prof. Samira Ebrahimi Kahou

University of Alberta Edmonton, Canada
Master of Science (Thesis) in Computing Science; ***GPA: 4.0/4.0*** *Sep 2017 – Sept 2019*
Advisor: Prof. Martha White
Thesis: Fixed Point Propagation: A New Way To Train Recurrent Neural Networks Using Auxiliary Variables

Jadavpur University Kolkata, India
Bachelor of Engineering in Electrical Engineering; ***GPA: 9.08/10.00*** *Aug 2013 – Jul 2017*

Research & Work Experience

Tata Consultancy Services, Research and Innovation Mumbai, India
Researcher; Data and Decision Sciences Team (Advisor: Dr. Harshad Khadilkar) *Nov 2019 – Dec 2021*

- Applied Reinforcement Learning techniques to solve multi-product, multi-node inventory management problem in Supply Chains, leading to ~20% improvement over current practices.
- Developed a generic Reinforcement Learning Framework for handling delayed actions and observations.

Indian Statistical Institute Kolkata, India
Research Intern; Computer Vision and Pattern Recognition Department *May – Jul 2015*

- Contributed to character segmentation methods for Bengali language using tesseract and cowboxer.
- Implemented an Optical Character Recognition for English Language with an accuracy of 95%.

Teaching Experience

University of Alberta Edmonton, Canada
Teaching Assistant; CMPUT 275, Introduction to Tangible Computing-II *Jan – April 2018*

- Instructed Lab Sessions & Graded Projects and Assignments for ~120 students

University of Alberta Edmonton, Canada
Teaching Assistant; CMPUT 274, Introduction to Tangible Computing-I *Sept – Dec 2017*

- Instructed Lab Sessions & Graded Projects and Assignments for ~150 students

Publications

1. **A Learning Based Framework for Handling Uncertain Lead Times in Multi-Product Inventory Management**
   European Workshops on Reinforcement Learning (EWRL) 2022
   Hardik Meisheri, **Somjit Nath**, Mayank Baranwal, Harshad Khadilkar
2. **Follow your Nose: Using General Value Functions for Directed Exploration in Reinforcement Learning**
   Reinforcement Learning in Games Workshop, AAAI 2022
   Somjit Nath, Omkar Shelke, Durgesh Kalwar, Hardik Meisheri, Harshad Khadilkar
3. **Revisiting State Augmentation methods for Reinforcement Learning with Stochastic Delays**
   Conference on Information and Knowledge Management (CIKM) 2021
   Somjit Nath, Mayank Baranwal and Harshad Khadilkar
4. **Scalable Multi-Product Inventory Control with Lead Time Constraints using Reinforcement Learning**
   Neural Computing and Applications Journal [Impact Factor = 5.102]
   Hardik Meisheri, Nazneen N Sultana, Mayank Baranwal, Vinita Baniwal, **Somjit Nath**, Satyam Verma, Balaraman Ravindran, Harshad Khadilkar

5. **SIBRE: Self Improvement Based REwards for Adaptive Feedback in Reinforcement Learning**
   International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2021
   Somjit Nath, Richa Verma, Abhik Ray, Harshad Khadilkar
6. **Training Recurrent Neural Networks Online by Learning Explicit State Variables**
   International Conference on Learning Representations (ICLR) 2020
   Somjit Nath, Vincent Liu, Alan Chan, Xin Li, Adam White, Martha White
7. **Two-Timescale Networks for Nonlinear Value Function Approximation**
   International Conference on Learning Representations (ICLR) 2019
   Wesley Chung, **Somjit Nath**, Ajin Joseph, Martha White
8. **A Fixed-Point Formulation for Recurrent Neural Networks**
   Continual Learning Workshop, NeurIPS 2018
   Somjit Nath, Taher Jafferjee and Martha White
9. **Rejection Sampling for Off-Policy Learning**
   Continual Learning Workshop, NeurIPS 2018
   Wesley Chung, Sina Ghiassian, **Somjit Nath** and Martha White
10. **Smartphone Camera Based Analysis of ELISA using Artificial Neural Network**
    IET Computer Vision Journal [Impact Factor = 1.95]
    Somjit Nath, Subhannita Sarcar, Biswendu Chatterjee, Nabendu Chatterjee, Rhishita Chourashi
11. **Arduino Based Door Unlocking System with Real Time Control**
    IEEE International Conference on Contemporary Computing and Informatics (IC3I) 2016
    Somjit Nath, Paramita Banerjee, Rathindra Nath Biswas, Swarup Kumar Mitra and Mrinal Kanti Naskar

Awards & Achievements

Ph.D. Fellowship: Awarded Ph.D. Fellowship to pursue Ph.D. in Engineering at École de technologie supérieure.

TCS Citation Award: Received the award **twice** (Feb 2022 & Oct 2021) for research contributions in terms of number of papers published.

Scholarship for Academic Excellence, State Electrical Engineers' Association: Awarded to undergraduate students who have been ranked in the top 3 students of their batch.

Runner-up in KSHITIJ, the technology fair of IIT Kharagpur: Participated and reached the Final of the Autonomous Robotics Event, "Sherlock" contested by ~30 teams.

Summer Fellow, Indian Academy of Sciences: Only person from Jadavpur University, Electrical Engineering Department to be selected for the year 2015.

Skills

Programming: Python, C++, C

Tools: Tensorflow, Pytorch, Jax, MATLAB, Octave, Arduino, Processing

Languages: English (Professional), Bengali (Native), Hindi (Professional)

Relevant Learning

Coursework: Introduction to Machine Learning (*CMPUT 551, Prof. Martha White*), Reinforcement Learning & AI (*CMPUT 603, Prof. Rich Sutton*), Optimization Principles in Reinforcement Learning (*CMPUT 659, Prof. Martha White*)

Summer School: Deep Learning & Reinforcement Learning Summer School, 2019

Other Interests

Sports: Table Tennis (Rating: 377). Played in Edmonton Chinatown Open 2018 & CUSTTA Open 2018, Calgary

Management: Organized hackathon 'SparkHack' and the circuit-building competition 'Circuistic' held at the annual department technology fair 'Convolution' of the Department of Electrical Engineering, Jadavpur University in 2016.

Mentoring: Mentored a group of 5 school students for inter-school Robotics Competition during 'Convolution' 2016.